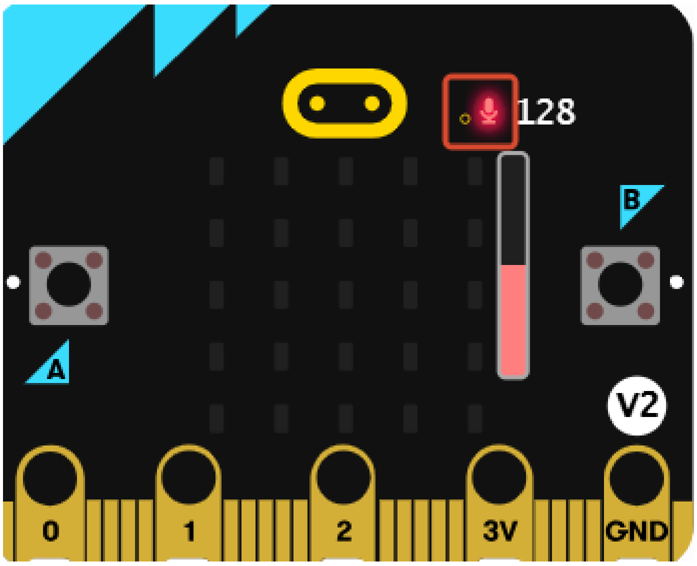
MIC Sound Intensity Collection V2

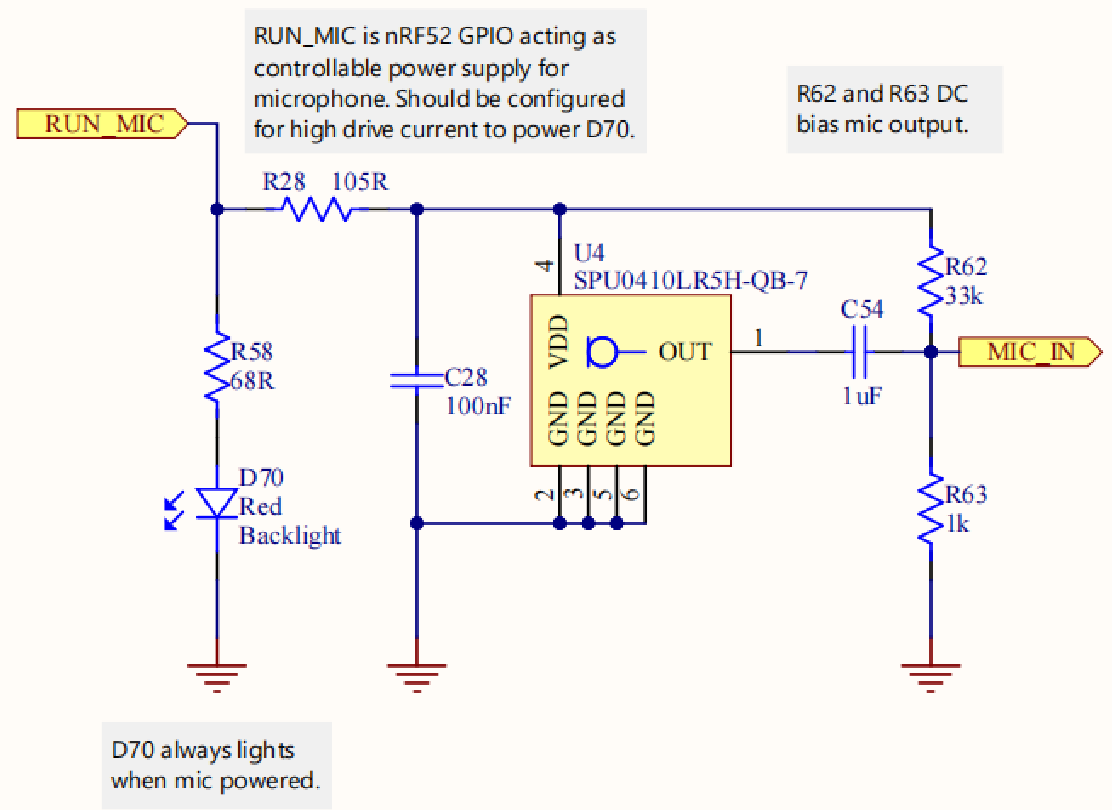
Experimental Purpose

The Micro:bit V2 main board is equipped with a sound sensor, located at the small hole in the red box. In this section, we will learn how to read the sound intensity information on the Micro:bit. The larger the value, the noisier the current environment.



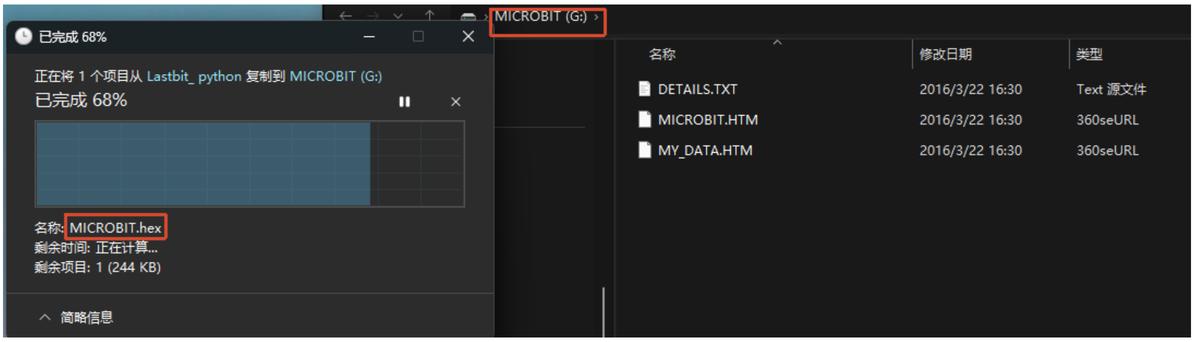
A microphone sensor is equipped near the Micro:bit Logo. When the program drives and reads it, the red LED will light up; when it is not being read, the red LED goes out and the microphone is turned off. This can reduce the power consumption of the Micro:bit at the hardware level.

Microphone



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**

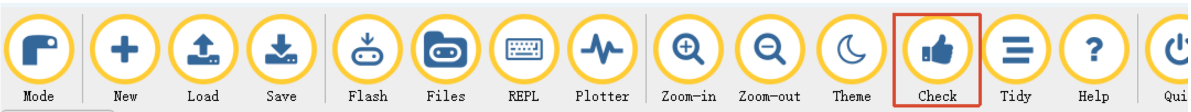
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

   Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.

6. Unplug the Microusb data cable and switch the switch to the ON position.

Code Implementation

```
# 6.2 MIC_Sound_Intensity_Collection_V2

from microbit import *

while True:

    # Read sound intensity
    volume = microphone.sound_level()
    display.scroll(volume)

    # Print sound intensity
```

```
    # print(str(volume) + '\n')
    sleep(100)
```

`microphone.sound_level()`: reads the sound intensity detected by the MIC. It returns integer data from 0 to 255.

Experimental Phenomenon

The Microbit LED matrix will display the sound intensity data obtained by the sound sensor.